

AC 230V Electromagnetic Relay

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An electromagnetic relay (EMR) allows a low-power circuit to switch on and switch off a relatively high-current circuit. Most of us are familiar with relays whose coils are designed to operate off supply voltage ranging from 5V to 48V. However, sometimes you want to use AC 230V EMRs for odd applications but are unsure about the circuit arrangement for doing the same. Besides, AC 230V EMRs are not within the reach of an average hobbyist. Here is a quick and affordable do-it-yourself solution.

Circuit and working

With the help of a suitable AC-DC step-down converter, you can make

your own compact and reliable AC 230V relay module.

First, select a PCB-mountable EMR and ensure that the relay coil has a very-low operating current. The author has used a type JQC-3FC (T73)-DC12V (from HL) in the prototype. The HL sugar cube relay draws around 25mA at 12V DC. Fig. 1 shows the author's prototype based on the basic circuit shown in Fig. 2.

There are several ways to convert an AC voltage into DC voltage required by the coil of the EMR. Traditionally, this has been done with a step-down transformer and a rectifier circuit.

However, in applications that involve providing a DC voltage to the relay and a few other low-current devices, transformer based power

supplies may not be so cost-effective. Fortunately, transformer-less power supplies provide a low-cost alternative to transformer based power supplies.

A capacitive transformer-less power supply is used in the circuit. The transformer-less power supply keeps the cost low and saves space in this low-current application. It also provides an energy-efficient solution.

The application is in two configurations, a basic circuit and an add-on circuit. These are shown in Figs 2 and 3, respectively.

The basic circuit is a transformer-less power supply for generation of 12V DC. It is built around a 5W, 100-ohm resistor (R1), 2.2µF, 400V capacitor (C1), 1N4007 diodes (D1 and

D2) and zener diodes 12V, 1W (ZD1 and ZD2) and a few other components.

The coil of 12V, 1C/O relay (RL1) is connected to CON3. Free-wheeling diode D3 is for protection of the coil. The relay circuit is connected to the basic circuit through connector CON2. The arrangement is shown in Fig. 2.

Resistor R1 and capacitor C1 (shunted by resistor R2) are connected in series to an AC source (CON1).

Remember, in order to get a DC voltage using this set up, a rectifier and a filter capacitor (for withstanding peak voltage in the set up) are used.

Warning. An electrocution hazard exists during experimentation with the transformer-less circuit that is

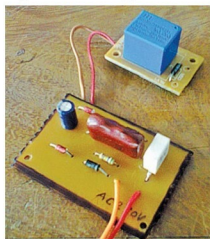


Fig. 1: Author's prototype

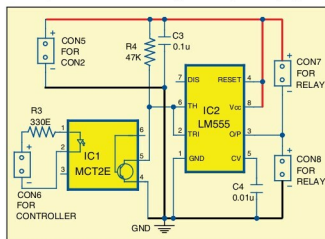


Fig. 3: Add-on relay driver circuit

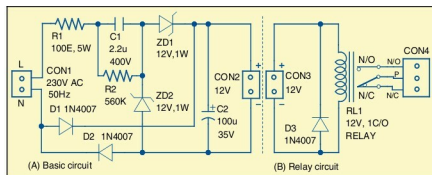


Fig. 2: Basic circuit with a relay